

SHIRA, Russian Federation

WMO: 297560

Lat: 54.500N

Lon: 89.933E

Elev: 474

StdP: 95.75

Time Zone: 7.00 (E07)

Period: 86-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.3	-30.1	-36.1	0.1	-33.0	-33.1	0.2	-30.0	10.2	-11.3	9.1	-10.3	0.4	100	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	28.8	17.7	26.7	17.1	24.8	16.2	19.0	25.3	18.1	24.4	17.2	23.1	3.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	12.5	21.5	15.7	11.8	21.0	14.8	11.2	20.3	56.0	25.4	52.9	24.1	50.2	23.0	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.9	6.8	-34.7	31.9	3.4	1.8	-37.2	33.1	-39.1	34.2	-41.0	35.2	-43.5	36.4		
(5)			-34.8	20.7	3.4	1.6	-37.3	21.9	-39.2	22.8	-41.1	23.7	-43.5	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-16.8	-12.8	-6.5	2.6	10.5	15.4	18.1	16.1	8.6	2.1	-8.8	-12.7	(6)
(7)		DBStd	13.34	8.07	6.83	7.08	4.85	5.71	4.07	3.18	3.45	4.25	4.84	7.74	6.98	(7)
(8)		HDD10.0	3874	832	639	512	224	68	7	0	2	75	248	564	703	(8)
(9)		HDD18.3	6256	1090	873	771	471	246	102	44	87	292	504	814	962	(9)
(10)		CDD10.0	731	0	0	0	3	84	170	251	189	33	1	0	0	(10)
(11)		CDD18.3	72	0	0	0	0	3	16	36	17	1	0	0	0	(11)
(12)		CDH23.3	738	0	0	0	0	79	158	299	195	6	0	0	0	(12)
(13)		CDH26.7	174	0	0	0	0	18	31	74	51	0	0	0	0	(13)
(14)	Wind	WSAvg	2.4	1.8	2.0	2.3	3.1	3.1	2.7	2.2	2.1	2.2	2.4	2.4	2.1	(14)
(15)	Precipitation	PrecAvg	758	39	34	50	60	63	76	84	82	67	75	69	60	(15)
(16)		PrecMax	977	123	90	136	122	146	134	169	161	121	159	137	139	(16)
(17)		PrecMin	585	1	1	14	15	11	28	50	19	15	30	36	11	(17)
(18)		PrecStd	102	26	21	31	27	34	24	27	28	28	31	25	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	3.8	12.0	20.1	29.3	30.0	31.2	30.5	23.9	16.5	8.7	4.4	(19)
(20)			MCWB	-1.7	-0.6	5.0	9.4	16.0	17.6	18.6	18.4	15.0	9.1	4.3	0.9	(20)
(21)		2%	DB	-2.2	0.9	8.0	17.0	25.4	27.2	28.9	27.8	20.3	13.8	5.5	0.9	(21)
(22)			MCWB	-4.6	-2.5	3.2	8.1	14.5	16.9	18.4	17.4	13.1	7.9	1.6	-2.1	(22)
(23)		5%	DB	-4.0	-1.0	5.3	14.1	23.0	25.1	26.7	25.3	17.8	11.7	3.2	-1.3	(23)
(24)			MCWB	-6.1	-3.7	1.0	6.7	13.0	15.6	17.8	16.9	11.6	6.9	-0.2	-3.9	(24)
(25)		10%	DB	-6.2	-3.0	3.0	11.1	20.6	23.1	24.7	22.8	15.7	9.5	1.1	-3.0	(25)
(26)			MCWB	-8.0	-5.2	-0.6	5.4	12.1	15.2	16.8	15.8	10.3	5.5	-1.7	-5.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-0.4	5.9	10.3	16.2	18.7	21.2	19.7	15.8	10.0	4.4	1.1	(27)
(28)			MCDWB	2.0	3.0	11.3	17.0	26.6	26.1	27.5	27.5	21.9	15.6	8.4	4.7	(28)
(29)		2%	WB	-4.5	-2.3	3.2	8.5	15.0	17.7	19.5	18.5	13.7	8.5	2.1	-1.7	(29)
(30)			MCDWB	-2.4	0.7	7.7	16.3	24.1	24.8	26.0	25.4	19.0	13.3	5.6	1.0	(30)
(31)		5%	WB	-6.1	-3.7	1.3	7.0	13.6	16.7	18.5	17.4	12.2	7.3	0.0	-3.8	(31)
(32)			MCDWB	-4.2	-1.1	5.1	13.3	21.7	23.1	24.3	24.1	16.7	11.5	2.8	-1.5	(32)
(33)		10%	WB	-8.0	-5.2	-0.5	5.5	12.4	15.7	17.6	16.3	10.9	5.7	-1.6	-5.3	(33)
(34)			MCDWB	-6.2	-2.9	2.6	11.0	19.8	21.8	23.2	22.1	14.7	9.4	1.1	-3.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	9.7	10.5	11.1	11.3	10.4	9.8	10.2	9.3	8.8	7.7	8.2	(35)
(36)			MCDBR	10.3	10.8	12.4	16.2	15.9	13.8	13.6	15.0	13.6	12.3	9.4	9.7	(36)
(37)			MCWBR	8.4	8.4	8.4	9.4	8.3	6.0	5.5	6.4	7.5	7.9	6.5	7.8	(37)
(38)		5% WB	MCDWB	10.8	10.7	11.9	14.3	14.0	12.4	10.9	12.5	12.1	11.6	9.5	9.5	(38)
(39)			MCWBR	8.7	8.5	8.0	8.8	7.7	5.7	5.2	5.8	7.2	7.8	6.9	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.264	0.297	0.347	0.393	0.400	0.397	0.416	0.398	0.359	0.335	0.299	0.265	(40)	
(41)		taud	2.090	2.057	2.010	2.141	2.205	2.311	2.273	2.292	2.345	2.279	2.116	2.029	(41)	
(42)		Ebn at Noon	663	766	806	826	844	850	824	815	803	724	602	556	(42)	
(43)		Edh at Noon	66	99	131	133	132	121	123	114	95	80	65	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.92	1.92	3.33	4.49	5.12	5.85	5.18	4.34	2.98	1.61	0.83	0.59	(44)	
(45)		RadStd	0.10	0.17	0.29	0.24	0.61	0.54	0.34	0.40	0.34	0.17	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page